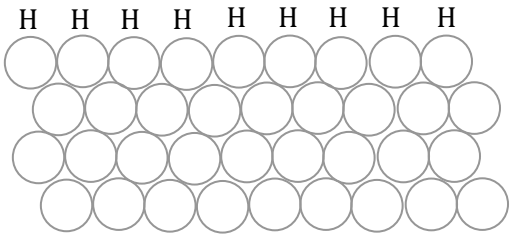
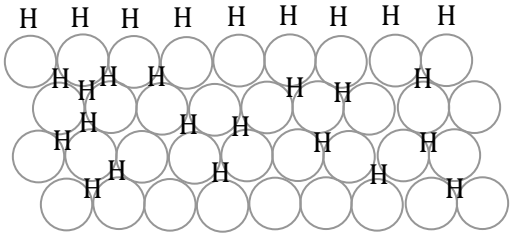
CEBT II REACTION ENGINEERING

4. Adsorption

- Adsorption is one key step of reaction on a surface.
- Adsorption is also a key part of many separations processes. Examples:
  - removing harmful organics from aqueous streams by adsorption in porous carbon
  - separating gas mixtures based on preferential adsorption of one component in a zeolite.
- Note that the word "adsorption" (with a "d") refers to a SURFACE process. The word "absorption" (with a "b") refers to a BULK process.



H_2 a**d**sorption
on palladium



H_2 a**b**sorption →
palladium hydride

4.1 Physical adsorption and chemical adsorption

- Physical adsorption (physisorption):
  - The adsorption is caused by weak Van der Waals interactions, and any Coulombic interactions that may be present (which could in principle include polar, dipolar and quadrupolar terms).
  - There is no significant redistribution of electron density in either the adsorbate molecule or at the substrate surface.
- Chemical adsorption (chemisorption):
  - The adsorption is caused by the formation of a chemical bond between the adsorbate and the substrate.
  - This must involve a substantial change in the distribution of electron density.
  - In some cases, the adsorbate is chemisorbed while still retaining its molecular identity (*molecular chemisorption*). An example is CO on Pd to give Pd–C≡O.
  - In other cases, formation of the bond(s) between adsorbate and substrate causes dissociation of the molecule to occur (*dissociative chemisorption*). An example is $H_2 \rightarrow 2H$ on Pd.

Property	Chemisorption	Physisorption
Saturation uptake	Limited to one monolayer	Multilayer adsorption possible
Nature of adsorption	May be dissociative May be irreversible	Non-dissociative Reversible
Enthalpy of adsorption (always negative)	Depends on strength of chemical bond formed (10–250 kJ mol^{-1})	Comparable with enthalpy change on condensation (5–30 kJ mol^{-1})
Kinetics of adsorption	Very variable – can be an activated process	Fast non-activated process
Adsorption temperature	Can be low or high temperature	Near or below T_{bp} of the gas (e.g. N_2<80 K, CO_2<200 K)
Crystallographic specificity	Different surfaces may behave differently	Independent of surface geometry

Notes:

1. For solids such as zeolites, the small pore sizes (of molecular dimensions) mean that a saturation loading is reached with physisorption before significant multilayer adsorption can take place.
2. Enthalpies of adsorption for physisorption depend on the size of molecule (as larger molecules have more interatomic contacts with the surface) and any dipolar interactions.

4.2 Amount adsorbed

- The amount of molecules that adsorb on a surface/particle may be measured experimentally, e.g. by gravimetric or volumetric measurements.
- The amount adsorbed is usually quoted in one of three ways:
  - as mass adsorbed per g of catalyst.
  - as the volume V_{STP} that the adsorbate molecules would occupy if they were in the gas phase at 273 K and 1 bar pressure (STP conditions).
  - as a *fractional coverage* θ, particularly if the number of adsorbed molecules reaches a limiting (saturation) value at high adsorbate pressures:

    $$\theta = \frac{\text{amount of adsorbed gas}}{\text{amount of adsorbed gas at saturation conditions}}$$

  - In the case of chemisorption the saturation coverage will be a monolayer, and the fractional coverage can alternatively be written:

    $$\theta = \frac{\text{number of surface sites occupied}}{\text{total number of surface sites}}$$

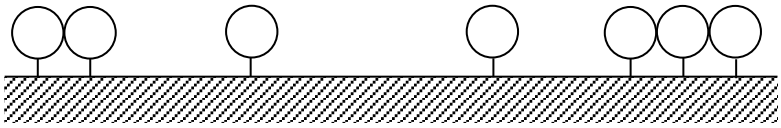
- An *adsorption isotherm* gives the relationship between amount adsorbed once equilibrium has been attained as the pressure (or concentration) is varied at a fixed temperature.

- An *adsorption isobar* gives the relationship between amount adsorbed once equilibrium has been attained as the temperature is varied at a fixed pressure.
- An *adsorption isostere* gives the relationship between the pressure and temperature at which equilibrium is attained for a fixed specified amount adsorbed.
- Isotherms, isobars and isosteres are time consuming to measure accurately. Usually isotherms are the easiest to measure.

4.3 Isotherms for molecular chemisorption of pure species

4.3.1 Langmuir adsorption isotherm

- The Langmuir isotherm is the simplest model for chemisorption.
- Assumptions:
  1. all possible sites for adsorption are equivalent (or behave equivalently).
  2. up to monolayer adsorption occurs; for chemisorption this is because a chemical bond needs to form between adsorbate and surface.
  3. the binding strength is independent of coverage; this requires that there are no interactions between neighbouring adsorbate molecules.



constant ΔH_{ads}

- At dynamic equilibrium the rate of adsorption of molecules will equal the rate of desorption of molecules from the surface.

Molecular adsorption

- Consider first the rate of adsorption of molecules.
- The number of gas-phase molecules striking a surface is proportional to the gas pressure.
- The probability that the gas-phase molecules strikes an empty site is $(1-\theta)$.
- We also need to allow for the fact that not every molecule striking an empty site will adsorb.
- The rate of adsorption is therefore

  $r_{\text{a}} = k_{\text{a}}(1-\theta)P$

  - r_{a} is sometimes expressed as moles/second; sometimes as moles/m^2/s; k_{a} needs appropriate units.
  - k_{a} will be independent of coverage if Langmuir's assumption of constant binding strength is correct.
  - For molecular adsorption k_{a} only has a small temperature dependence that is usually ignored.

- The rate of desorption is expected to be proportional to the number of adsorbed molecules.

  $r_d = k_d\theta$

  - r_d is sometimes expressed as moles/second; sometimes as moles/m^2/s; k_d needs appropriate units.
  - k_d is expected to be have a strong temperature dependence; an activation energy needs to be overcome for desorption to occur.
- At dynamic equilibrium

  $r_a = r_d$

  $k_a(1-\theta)P = k_d\theta$
- Let $K = k_a/k_d$ so

  $KP(1-\theta) = \theta$

  $\theta = \dfrac{KP}{1+KP}$ [Equation for Langmuir's adsorption isotherm]
- K will not depend on coverage θ if Langmuir's assumption of constant binding strength is obeyed.

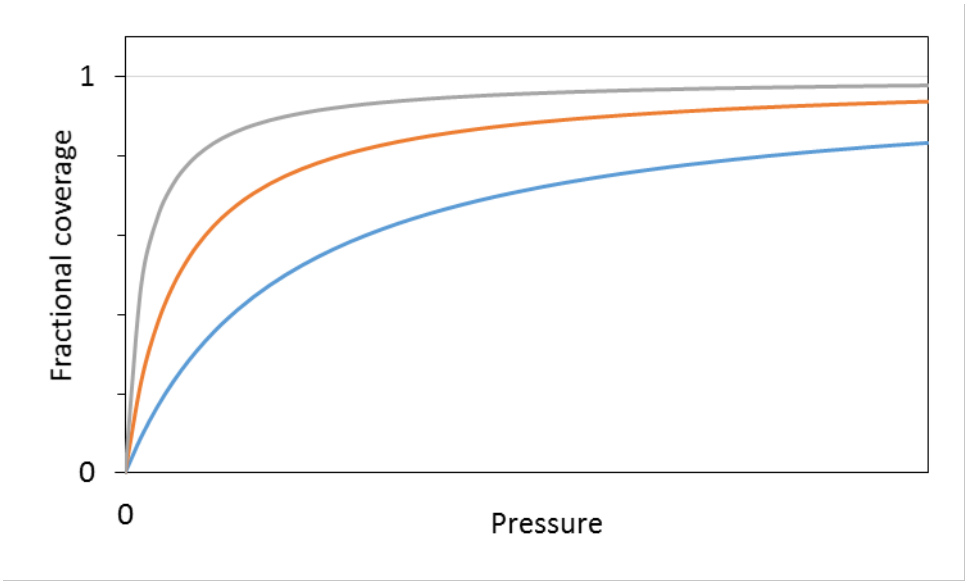


Figure: Examples of the Langmuir isotherm at three different temperatures

- K is the equilibrium constant for the adsorption reaction $A_{(g)} + * \rightleftharpoons A_{(*)}$ where * denotes a surface site. It will have a temperature dependence given by the van't Hoff equation

  $\dfrac{d\ln K}{dT} = \dfrac{\Delta H_{ads}}{RT^2}$

  with ΔH_{ads} independent of θ with Langmuir's assumption.
  - In the special case that ΔH_{ads} is approximately independent of T we get

    $K = K_o \exp\left(\dfrac{-\Delta H_{ads}}{RT}\right)$

- Note that adsorption is always exothermic, so ΔH_{ads} is negative.
- As T increases K will therefore decrease, and so the amount adsorbed at equilibium will decrease as expected.

- From $K = k_a/k_d$ we expect the "activation energy" for K will be $E_a - E_d$.
  - Non-activated adsorption, $E_a = 0$, is common for molecular adsorption (i.e. adsorption without dissociation). In this case $\Delta H_{ads} = -E_d$, so the enthalpy of adsorption is numerically equal to the activation energy for desorption.
- The Langmuir equation is only rarely obeyed exactly – its assumptions are not generally true.
- However, it continues to be widely used because of its simplicity and the fact that it is capable of predicting many observed phenomena in heterogeneous catalysis and gas separations.
- Other isotherms that are commonly used for chemisorption are given in the following table where q denotes the amount adsorbed (rather than fractional coverage).

		Adjustable parameters	Comments
Henry	$q = K_H P$	1	only valid at very dilute loadings
Langmuir	$q = \frac{q_{sat} K P}{1 + K P}$	2	ΔH_{ads} is independent of coverage
Freundlich	$q = K P^{1/n}$	2	ΔH_{ads} is a logarithmic function of coverage
Temkin	$q = K_1 \ln(K_2 P)$	2	ΔH_{ads} is a linear function of coverage
Toth	$q = \frac{q_{sat} K P}{[1 + (K P)^t]^{1/t}}$	3	allows for a distribution of ΔH_{ads} values (i.e. sites can have different adsorption strengths)
Others	…	2, 3 or more	Various advantages and disadvantages

- Notes:
  - For adsorption from the liquid phase, replace pressure by concentration.
  - The Freundlich and Temkin isotherms are unsuitable at very high pressures (as they incorrectly predict $q \to \infty$ as $P \to \infty$) and at very low pressures (as they do not predict linear behaviour as $P \to 0$).
  - Determination of q_{sat} using Langmuir or Toth enables the number of surface sites on which adsorption can occur to be determined. This is often then converted into the surface area available for adsorption.

4.4 Isotherms for dissociative chemisorption

- Example: H_2 molecules dissociate into two H atoms when adsorbing on a metal.

  $H_{2(g)} + * + * \longrightarrow 2H_{(*)}$

  - Here * denotes a vacant surface site.
- Rate of H_2 adsorption

  $r_a = k_a(1-\theta)^2 P$

  because there need to be two neighbouring vacant metal sites for the dissociation to occur.
- Rate of H_2 desorption

  $r_d = k_d\theta^2$
- At dynamic equilibrium

  $k_a(1-\theta)^2 P = k_d\theta^2$
- With $K = k_a/k_d$ we get

  $$\theta = \frac{\sqrt{KP}}{1+\sqrt{KP}}$$

4.5 Isotherms for adsorption of mixtures

- If the adsorbed species compete for the same sites ("competitive adsorption), then we can extend Langmuir's treatment to consider mixtures.
- Consider non-dissociative adsorption of A and B on the same sites.
  - Let θ_* be the fraction of vacant sites on the surface
  - Then

    $k_{a,A}P_A\theta_* = k_{d,A}\theta_A \quad \text{so} \quad \theta_A = K_A P_A \theta_*$

    $k_{a,B}P_B\theta_* = k_{d,B}\theta_B \quad \text{so} \quad \theta_B = K_B P_B \theta_*$
  - Adding

    $\theta_A + \theta_B = K_A P_A \theta_* + K_B P_B \theta_*$
  - Hence

    $1 - \theta_* = (K_A P_A + K_B P_B)\theta_*$

    $$\theta_* = \frac{1}{(1 + K_A P_A + K_B P_B)}$$
  - The isotherms are therefore

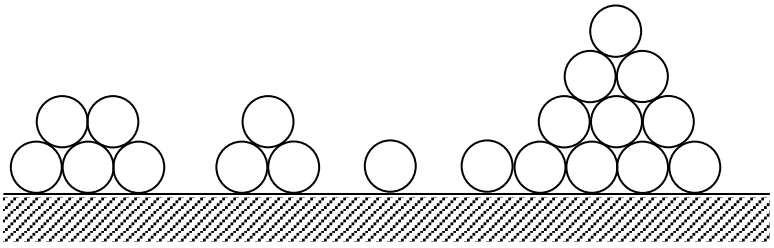
    $$\theta_A = \frac{K_A P_A}{(1 + K_A P_A + K_B P_B)}$$

    $$\theta_B = \frac{K_B P_B}{(1 + K_A P_A + K_B P_B)}$$
- Similar expressions can be obtained that allow for dissociation of species with competitive adsorption.

- While not normally the case on metals, on metal oxides we can find the case of non-competitive adsorption: species A may adsorb on one type of site (e.g. at the metal), while species B adsorbs on another (e.g. at an oxygen vacancy).

4.6 Isotherms for physisorption

- The simplest isotherm used to describe physisorption is the so-called BET isotherm (after Brunauer, Emmett and Teller).
- The BET model allows multilayer adsorption to occur. The assumptions are
  1. All possible sites for adsorption on the solid surface are equivalent and binding strength is independent of coverage. [These are the same assumptions as the Langmuir model].
     - Let the adsorption equilibrium constant for molecules in layer 1 be K_1. It reflects the interactions between adsorbate molecules and the solid surface.
  2. All possible sites for adsorption in higher layers are equivalent and binding strength is independent of coverage.
     - Let the adsorption equilibrium constant for molecules in layer 2 be K_2. It reflects the interactions between adsorbate molecules and adsorbate molecules.
     - It is assumed that the adsorption equilibrium constant for molecules in higher layers is identical to K_2.



  3. The amount adsorbed tends to infinity as P tends to P^{sat}.
     - We know that vapour condenses to liquid at P^{sat}, and so we therefore expect an infinite number of layers to adsorb on a surface at this pressure.
- With these assumptions, the derivation of the BET isotherm is not hard, but it is slightly awkward because it involves sums of infinite series and so only the result is quoted here.

- The BET isotherm is normally expressed as (Data Book page 33)

$$\frac{P}{V(P^{\text{sat}} - P)} = \frac{1}{V_{\text{m}}C} + \frac{C-1}{V_{\text{m}}C}\frac{P}{P^{\text{sat}}}$$

Here:

- V is the amount of gas adsorbed at pressure P (usually converted to the equivalent volume of gas at STP conditions)
- V_{m} is the amount of gas adsorbed that corresponds to a monolayer of adsorbate
- P^{sat} is the saturated vapour pressure at the temperature of the experiment
- C is a constant which equals the ratio of equilibrium constants, $C = K_1/K_2$

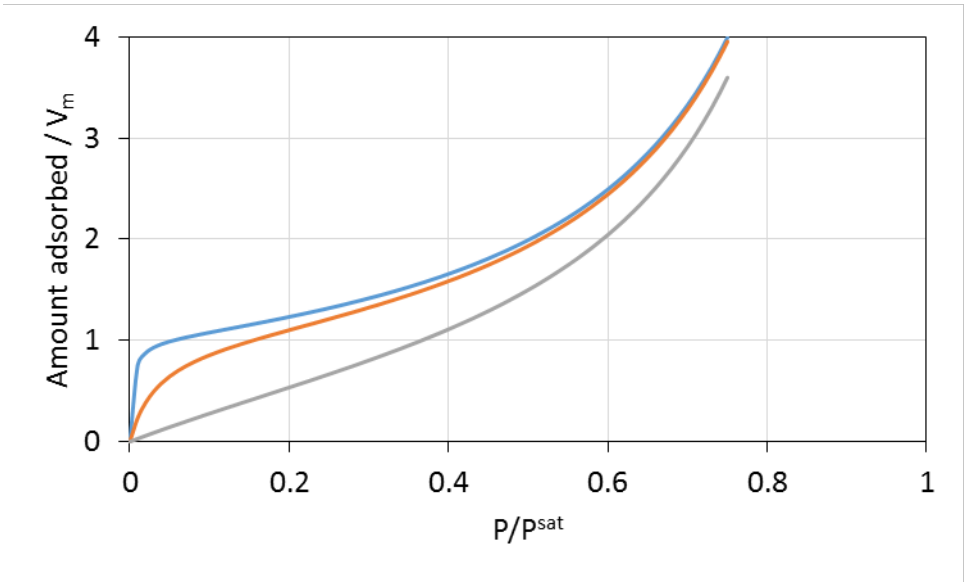


Figure: BET isotherm predictions for high C, medium C and low C values

- The BET parameter $C = K_1/K_2 = (k_{\text{a1}}/k_{\text{d1}})/(k_{\text{a2}}/k_{\text{d2}})$. If the pre-exponential factors for adsorption and desorption are the same for layer 1 as the higher layers, then this means

$$C = \exp\left(\frac{-(\Delta H_{\text{ads1}} - \Delta H_{\text{ads2}})}{RT}\right)$$

where ΔH_{ads1} is the enthalpy of adsorption on layer 1, and ΔH_{ads2} is the enthlpy of adsorption on subsequent layers (both are negative numbers).

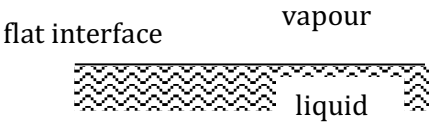
- If the adsorption strength for layer 1 is significantly greater than for subsequent layers, then $C \gg 1$. This is common in practice.
- If the adsorption strength for layer 1 is comparable to subsequent layers then C will be close to 1. This is rare.

- It is possible in principle to estimate a value of C by measuring ΔH_{ads1} and then taking ΔH_{ads2} to be the enthalpy of condensation from vapour to liquid (or solid). However, this method rarely fits experimental data well, and so C is almost always treated as an adjustable parameter.

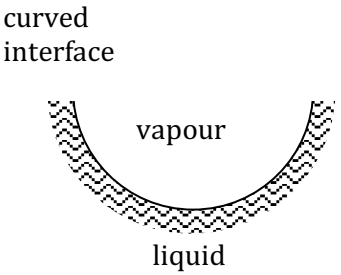
- For most physisorption systems, the BET equation is reasonably accurate in the pressure range $0.05 < P/P^{\text{sat}} < 0.35$
  - Experimental data often fit the form of the isotherm well in this range, even if some of the assumptions in its derivation are not exactly obeyed.
  - A linear plot can be constructed by plotting $P/V(P^{\text{sat}} - P)$ against P/P^{sat}.
    - Parameter C can be found from gradient / intercept, and then parameter V_{m} can be found from the gradient.
    - Sometimes C is assumed to be large, in which case the gradient is $1/V_{\text{m}}$.
  - Once V_{m} is known, then the surface area of the sample can be calculated.

4.7 Capillary condensation within porous solids

- When P/P^{sat} exceeds about 0.4, a phenomenon called capillary condensation may take place when measuring isotherms of mesoporous solids, namely those with pores in the range 2 – 50 nm.
- The T, P conditions at which the vapour → liquid transition occurs for a pure fluid depends on the curvature of the liquid-vapour interface.



T fixed: VLE when $P = P^{\text{sat}}$



T fixed: VLE when $P = P^{\text{cond}}$

with $P^{\text{cond}} < P^{\text{sat}}$

- This is a consequence of surface tension, in the same way that the vapour pressure of a small droplet is elevated above the saturated vapour pressure as a consequence of surface tension raising the pressure within the droplet (but the curvature is the other way around in this case, so the vapour pressure is less than the saturated vapour pressure).

- The condensation pressure P^{cond} at which the vapour → liquid transition is in equilibrium is given by the Kelvin equation (link CEBT II Thermodynamics: derivation of vapour pressure exerted by a droplet).

$$\ln\left(\frac{P^{\text{cond}}}{P^{\text{sat}}}\right) = -\frac{2\sigma \underline{V}^{\text{liq}} \cos\theta}{RT}\frac{1}{r_{\text{curv}}}$$

where

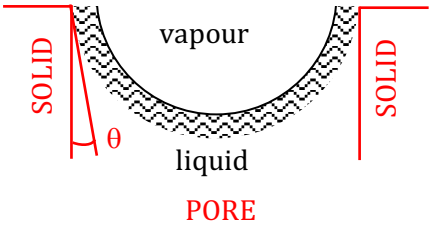
r_{curv} is the mean radius of curvature

σ is surface tension of the liquid

$\underline{V}^{\text{liq}}$ is the molar volume of the liquid

θ is the contact angle between liquid and solid (this term is not present for droplets or bubbles).

curved interface



  - The mean radius of curvature is given by

$$\frac{1}{r_{\text{curv}}} = \left(\frac{1}{r_1} + \frac{1}{r_2}\right)/2$$

where r_1 and r_2 are the radii of curvature in principal directions (i.e. pores need not be of circular cross section, unlike droplets/bubbles).

- Physisorbed molecules on the surface behave as "liquid" in the diagram above.
- In the particular case of adsorbed nitrogen at 77 K:
  - σ and $\underline{V}^{\text{liq}}$ are known.
  - The contact angle θ is not known and it is hard to measure. It is usually assumed that θ is close to zero in which case $\cos\theta \sim 1$.
  - In this particular case, the Kelvin equation gives:

$$\ln\left(\frac{P^{\text{cond}}}{P^{\text{sat}}}\right) = -\frac{0.959\ \text{nm}}{r_{\text{curv}}}$$

- For an adsorbate within nanometre pores, the Kelvin equation predicts that vapour → liquid condensation will occur in pores at a lower pressure than P^{sat}.

- For pores of molecular dimensions (say $r < 1$ nm), the pores are so small that a "bulk" vapour → liquid transition cannot occur.
- For macropores ($r > 50$ nm), the pores are so large that there is negligible change in the vapour → liquid transition pressure.
- For mesopores (1 nm < r < 50 nm), capillary condensation occurs: the vapour → liquid transition occurs within the pores at a significantly lower pressure than P^{sat}.

- The Kelvin equation implies large **<u>negative</u>** pressure in the liquid phase for small pores:

$$\Delta P = \frac{2\sigma}{r}\cos\theta$$

with the pressure in the liquid being lower than in the vapour (i.e. the opposite of droplets).

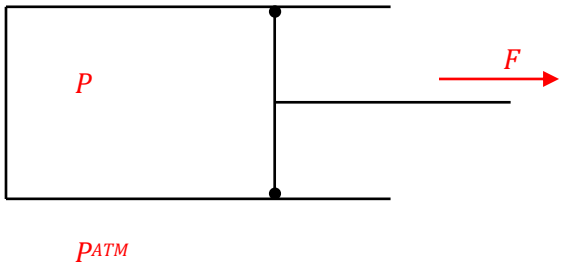
Say $\sigma = 0.07\ \text{Nm}^{-1}$ (value for water in contact with air at room temperature)

and $r = 14\ \text{nm} = 1.4 \times 10^{-8}\ \text{m}$

$\Delta P = 10^7 \cos\theta\ \text{Pa} \approx 100\ \text{bar}$

Vapour is usually at less than 1 atm, so liquid is at around −100 bar!

This **<u>is</u>** possible. Consider a smooth-walled, liquid filled piston, of cross-sectional area, A:



$$P = P^{ATM} - \frac{F}{A}$$

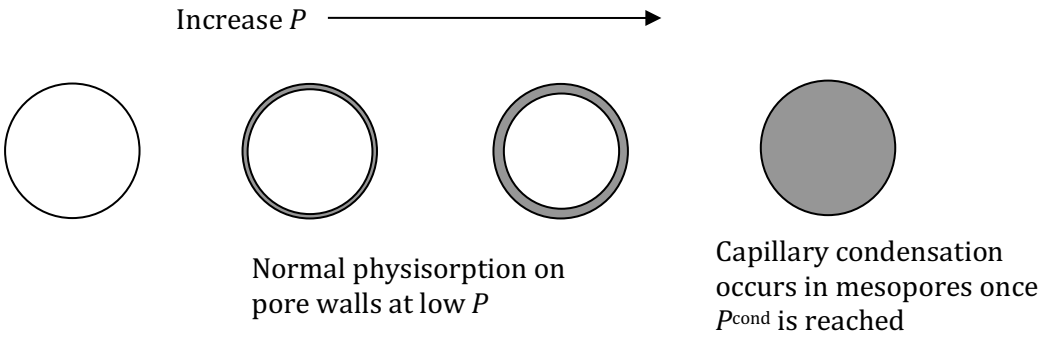
which can easily be negative.

The liquid is held together by cohesive forces and to the piston walls by adhesive forces (which can be large, e.g hydrogen bonds). Nucleation is difficult because the walls are smooth (remember bubbles and super-heated liquid from Equilibrium Thermo), so vapour cannot easily form. Hence the liquid is meta-stable at negative pressure.

- In the case of pores, we have genuine equilibrium because the curved interface results in mechanical equilibrium with a lower pressure in the liquid than the vapour. A bubble at equilibrium with the condensed liquid in a pore would have a diameter at least as large as the pore diameter, and so cannot form.
- The adhesive forces involved can be so large that the solid structure is destroyed when capillary condensation occurs.

4.7.1 Hysteresis when performing adsorption and desorption experiments

- Consider increasing the pressure for physisorption within a cylindrical mesopore:

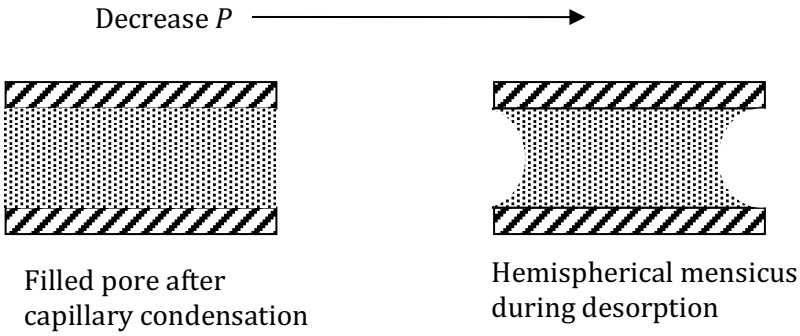


- Capillary condensation means that more adsorption will suddenly occur when P^{cond} is reached compared to that expected if adsorption proceeded solely by a multilayer adsorption mechanism.
- For adsorption in pores that are uniform cylinders,

  $r_1 = r_{pore} - t$; $r_2 = \infty$

  $r_{curv} = 2(r_{pore} - t)$

  where r_{pore} is the pore radius and t is the thickness of any adsorbed film.
- The pressure at which capillary condensation occurs can be used to calculate the radius of curvature r_{curv} by the Kelvin equation and thus r_{pore} (after using a physisorption isotherm model to estimate t).
- Now consider decreasing the pressure after capillary condensation has taken place within a cylindrical mesopore:



- In this case, there is a hemispherical meniscus during desorption

  $r_1 = r_{pore} - t$; $r_2 = r_{pore} - t$
- The mean radius of curvature is

  $r_{curv} = (r_{pore} - t)$

  where r_{pore} is the pore radius and t is the thickness of any adsorbed film.

- Hence the pressure at which capillary condensation occurs is different during a desorption experiment than in an adsorption experiment.
- This phenomenon is termed hysteresis.
  - For porous solids, there is often more than one state at which adsorbate molecules are at local thermodynamic equilibrium (i.e. a metastable state) rather than global equilibrium.

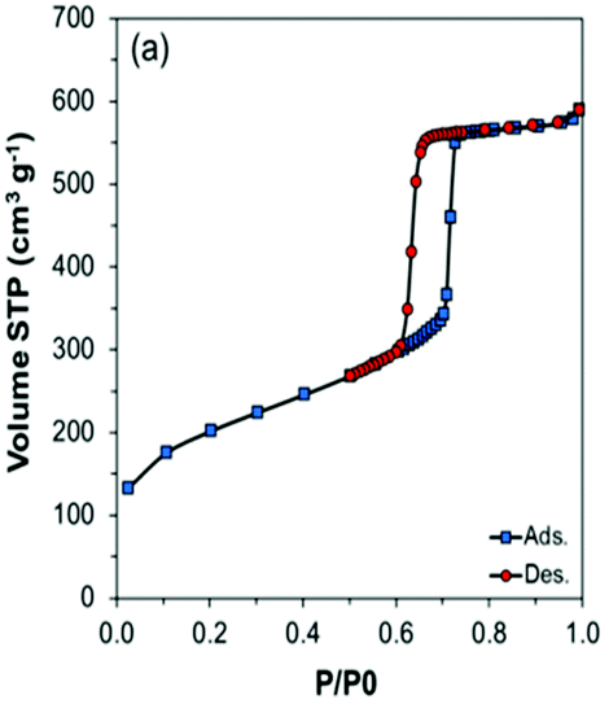


Figure: Experimental adsorption isotherm for N_2 in SBA-15, a mesoporous silica with cylindrical pores (from Cychosz et al., Chem. Soc. Rev. ***46****, 389 (2017).*

- As capillary condensation takes place in mesopores, the occurrence of hysteresis in an adsorption isotherm is a useful qualitative indication that mesopores are present.
- The exact shape of the hysteresis loop depends on the pore properties. It may be modelled to get information on pore shape and geometry.
  - For example, different shape hysteresis loops are predicted for slit-shaped pores, for cylindrical pores, and for "bottleneck" pores (i.e. large pores that have narrow pore openings).
  - The best way of characterizing the pore shapes and sizes that are present in a porous solid remains an active research area.

4.8 Obtaining enthalpies of adsorption from equilibrium adsorption data

- Enthalpies of adsorption are useful for characterizing the binding strength between adsorbate and solid.
  - They can be obtained calorimetrically (e.g. by measuring temperature increase upon adsorption)
  - They can also be obtained from equilibrium adsorption data as outlined here.
- If the Langmuir isotherm assumptions are obeyed exactly then ΔH_{ads} will be independent of coverage and

$$\theta = \frac{KP}{1 + KP}$$

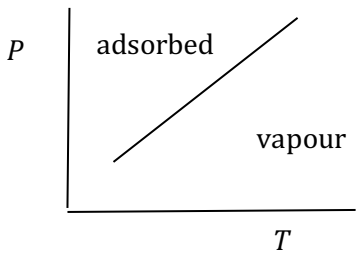
  - K is an equilibrium constant and so obeys the van't Hoff equation

$$\frac{d \ln K}{dT} = \frac{\Delta H_{\text{ads}}}{RT^2}$$

  - If K is measured at several temperatures, a plot of $\ln K$ vs $1/T$ will give a straight line from which ΔH_{ads} can be obtained.
- However, in many cases ΔH_{ads} is dependent on coverage and the Langmuir assumptions are not valid.
- For dilute systems ($\theta \rightarrow 0$), all systems obey Henry's law even if they do not obey the Langmuir isotherm. The Henry's law constant K_{H} obeys the van't Hoff equation

$$\frac{d \ln K_{\text{H}}}{dT} = \frac{\Delta H_{\text{ads},\,\theta \rightarrow 0}}{RT^2}$$

  - If K_{H} is measured at several temperatures, the enthalpy of adsorption in the limit of zero coverage can be obtained.
- For adsorption of gases in non-dilute systems:
  - we expect the enthalpy of adsorption will depend on coverage;
  - we can use the Clausius-Clapeyron equation for phase equilibrium.



$$\left(\frac{\partial \ln P}{\partial T}\right)_{\text{phase change}} = \frac{\Delta H_{\text{ads}\rightarrow\text{vap}}}{RT^2}$$

- The isosteric enthalpy of adsorption is therefore

$$\Delta H_{\text{ads},\,\theta} = -RT^2 \left(\frac{\partial \ln P}{\partial T}\right)_\theta$$

$$\Delta H_{\text{ads},\,\theta} = R \left(\frac{\partial \ln P}{\partial (1/T)}\right)_\theta$$